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FNCE30007 Practice Midsemester 2025 with coversheet

Derivative Securities (University of Melbourne)



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Student number:

Practice Mid-Semester Test, Sem 1
2025

Finance

FNCE30007 Derivative Securities

Reading time: 10 minutes

Writing time: 1 hour

Permitted Materials

- Calculator (Casio FX82 only);

Instructions to Students

During writing time, clearly write your student number on the cover sheet in the above student number box only.

Ensure each digit is clear.

Write your multiple choice answers in capital letters in the designated answer boxes provided within the exam paper for each question.

Do not communicate with other students in the exam venue.

No part of the exam paper may be removed from the venue.

Ensure you only have standard permitted materials on your desk plus additional materials if listed above.

Raise your hand if you require assistance.

During reading time, students may not write.

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There are fifteen multiple choice questions. Answer all questions. Choose the best and most complete answer to the question that is asked by writing your CAPITAL letter answer in the box underneath each question. Each question is worth three marks for a total of 45 marks.

1. An investor enters into a long position in 50 futures contracts. Each contract is for 1,000 units of the underlying asset and the contract price is \$5.50 per unit. They later close out the position at a contract price of \$6.50 per unit. What is the profit or loss on the position to the nearest dollar?

   A. -\$50,000
   B. \$50,000
   C. \$1,000
   D. -\$50
   E. None of the other answers.

   \$500
   650

   PnL = $(6.5 - 5.5) \times 1000 \times 50 = \$50000$

   Write your CAPITAL letter answer in this box: C B

2. Which of the following statements about derivatives is/are true?

   i. Because derivatives markets are small relative to other asset classes, they are unlikely to pose systemic risk for financial stability. ?
   ii. Significant derivatives trading volume takes place both on exchanges and in over-the-counter markets.
   iii. Margining (futures) only takes place in exchange-traded derivatives trades and not in OTC trades
   iv. Central Clearing Counterparties are institutions that intermediates derivative trades by taking settlement responsibility for each side of a trade. This helps simplify the risk management for individual traders.

   A. ii. only
   B. iii. only
   C. ii. and iv. only
   D. i., ii., iii., and iv.
   E. None of the other answers.

   Write your CAPITAL letter answer in this box: D C

3. A company enters into a long futures contract to buy 1,000 units of a commodity for \$60 per unit. The initial margin is \$6,000 and the maintenance margin is \$4,000. What futures price will allow \$2,000 to be withdrawn from the margin account?

   A. \$58
   B. \$62
   C. \$64

   $1000P = 60(1000) + 4000$

   $P = 64$

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$(F - 60) \times 1000 = 2000$

$F = 62$

D. \$66

E. None of the other answers.

Write your CAPITAL letter answer in this box: C B

4. A natural gas producer is exposed to the price of liquefied natural gas (LNG). They are considering hedging this exposure via crude oil futures. Each futures contract is for 1,000 barrels of oil. The firm has estimated the standard deviation of price changes for LNG is $\sigma_s = 0.065$ and for crude oil futures is $\sigma_F = 0.06$. The correlation coefficient in price changes is $\rho = 0.80$. What is the minimum variance hedge ratio (MVHR) and corresponding number of futures contracts to hedge an exposure of 3,500 barrels of LNG?

A. The MVHR is 0.867 and the optimal number of contracts is 3.

B. The MVHR is 0.939 and the optimal number of contracts is 3.

C. The MVHR is 0.867 and the optimal number of contracts is 0.25.

D. The MVHR is 1 and the optimal number of contracts is 1.

E. None of the other answers.

$h^{min} = 0.8\left(\frac{0.065}{0.06}\right)$

$= 0.867$

$\frac{0.867\ (3500)}{1000} = 3.0345$ contracts

Write your CAPITAL letter answer in this box: A

5. A natural gas producer is exposed to the price of liquefied natural gas (LNG). Which of the following derivative positions would fix a "worst-case" price the company will receive for future LNG production??

i. Long LNG call

ii. Short LNG call

iii. Long LNG put

iv. Short LNG put

v. Long LNG futures

vi. Short LNG futures

Producer : fall : long put

short futures

A. ii., iii. and v. only

B. i. and vi. only

C. iv. and v. only

D. iii. and vi. only

E. None of the other answers.

Write your CAPITAL letter answer in this box: D

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6. On March 1 a commodity's spot price is \$60 and its August futures price is \$59. On July 1 the spot price is \$64 and the August futures price is \$63.50. On March 1, a company entered into an August futures contracts to hedge its purchase of the commodity on July 1. It closed out the hedge position on July 1 and bought the commodity in the spot market. What is the effective price (after taking account of hedging) paid by the company (to 2 decimal places)?

Cost/value of hedged asset $= F_0 + (S_T - F_T)$
$= 59 + 64 - 63.50$
$= \$59.5$

A. \$59.50
B. \$60.50
C. \$61.50
D. \$63.50
E. None of the other answers.

Write your CAPITAL letter answer in this box: A

7. Which of the following is/are examples of basis risk?

i. Hedging an exposure to the price of platinum using gold futures
ii. Expressing a bullish view on the Australian equity market with SPI/ASX 200 futures
iii. Taking offsetting long and short positions in 3 year and 10 year government bond futures.
iv. Hedging an exposure to the price of crude oil on 15 April with a futures contract expiring 30 April

A. i. only
B. ii. only
C. i., ii., iii. and iv.
D. iv. only.
E. None of the other answers.

Write your CAPITAL letter answer in this box: D E

8. Consider a continuously compounded interest rate of 8%pa. Answering in percent to two decimal places, the equivalent discrete annualised interest rate paid monthly is:

$r^m = 12 \times (e^{\frac{8\%}{12}} - 1)$
$= 8.03\%$

A. 0.67%pa
B. 100%pa
C. 7.97%pa
D. 8.03%pa
E. None of the other answers.

Write your CAPITAL letter answer in this box: D

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discrete - dividend paying asset

9. The spot price of an investment asset is \$30 and the continuously compounding risk-free rate for all maturities is 10%pa. The asset provides an income of \$2 at the end of the first year and at the end of the second year. What is the no-arbitrage price of a three-year forward price (to 2 decimal places)? You may assume transaction costs and other frictions are absent when answering this question.

$F_0 = (S_0 - I)e^{rT}$ $\quad I$ = dividend

$= (30 - 3.4471)e^{10\%(3)}$ $\quad = 2e^{-10\%} + 2e^{-10\%(2)}$

$= 35.84$ $\quad = 3.4471$

A. \$40.50

B. \$35.84

C. \$19.67

D. \$22.22

E. None of the other answers.

Write your CAPITAL letter answer in this box: B

10. What should an arbitrageur do when the one-year forward price of an asset is too high? Assume that the asset provides no income.

A. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a long forward contract to buy the asset in one year. ✗

B. The arbitrageur should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a short forward contract to sell the asset in one year.

C. The arbitrageur should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a long forward contract to buy the asset in one year.

D. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a short forward contract to sell the asset in one year.

E. None of the other answers.

Write your CAPITAL letter answer in this box: D

11. Suppose that the continuously compounded risk-free rate is 10%pa with continuous compounding and that the continuously compounded dividend yield on a stock index is 4%pa. The index price is 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

$F_0 = S_0 e^{(r-d)T} = 400\, e^{(10\% - 4\%)\left(\frac{4}{12}\right)} = 408.08$ (theoretical futures price)

A. Go long the futures contract, borrow ✗ at the risk-free rate and buy the stocks underlying the index to lock in a profit of 3.08 in four months time.

B. Go short the futures contract, borrow at the risk-free rate and buy the stocks underlying the index to lock in a profit of 3.08 in four months time. ✗

C. Go long the futures contract, short-sell the stock underlying the index and deposit the proceeds at the risk-free rate to lock in a profit of 3.08 in four months time.

D. Go long the futures contract, short-sell the stock underlying the index and deposit the proceeds at the risk-free rate to lock in a profit of 1.08 in four months time. ✗

E. None of the other answers.

- long index futures

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Write your CAPITAL letter answer in this box: C

12. Which of the following is NOT true about call and/or put options:

A. The price of a call option increases as the strike price increases ∝ spot
B. An American option can be exercised at any time during its life ✓
C. A European option can only be exercised on the maturity date ✓
D. Investors must pay an upfront price (the option premium) for an option contract ✓
E. None of the other answers.

Write your CAPITAL letter answer in this box: A

13. The price of a stock on July 1 is \$57. A trader buys 100 call options on the stock with a strike price of \$60 when the option price is \$2. The options are exercised when the stock price is \$65. The trader's net profit is:

A. \$700
B. \$500
C. \$300
D. \$600
E. None of the other answers.

* This is not forward/futures, the strike price represents the stock price and not the price you're buying at!

$(65 - 60) - 2 = 3$

$3 \times 100 = 300$

Write your CAPITAL letter answer in this box: C

14. When interest rates increase with all else remaining the same, which of the following is true regarding European options? exercise at maturity

A. Both calls and puts increase in value
B. Both calls and puts decrease in value
C. Calls increase in value while puts decrease in value
D. Puts increase in value while calls decrease in value
E. None of the other answers.

call value ↑
put value ↓

because higher IR = ↑ discount rate = ↓ PV

Write your CAPITAL letter answer in this box: C

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15. The price of a European put with a strike price of \$10 that expires in four months is \$0.50. The underlying stock price is \$11 and the stock pays a dividend of \$0.30 in three months time. The continuously compounded risk-free interest rate for all maturities is 6%pa. Assuming no transaction costs or other frictions, what is the no-arbitrage price of a European call option that expires in four months and also has a strike price of \$10? Answer to the nearest cent.

$c + 0.3e^{-6\%\left(\frac{3}{12}\right)} + 10e^{-6\%\left(\frac{4}{12}\right)} = 0.5 + 11$

$c = 1.40$

A. \$1.40

B. \$0.20

C. \$0.55

D. \$3.15

E. None of the other answers.

Write your CAPITAL letter answer in this box: A

END OF EXAM

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FORMULA SHEET

Cost/value of hedged asset $= F_0 + (S_T - F_T) = F_0 + b_T$

$F_0 + (S_T^* - F_T) + (S_T - S_T^*)$

$h^{min} = \rho \dfrac{\sigma_s}{\sigma_{F^*}}$

$N^{min} = h^{min} \dfrac{Q_A}{Q_F}$ or $N^{min} = \beta \dfrac{V_A}{V_F}$

$N^* = (\beta - \tilde{\beta}) \dfrac{V_A}{V_F}$

$r^* = m \ln \left(1 + \dfrac{r^m}{m}\right)$

$r^m = m \times \left(e^{r^*/m} - 1\right)$

$F_0 = S_0 e^{rT}$

$F_0 = (S_0 - I) e^{rT}$

$F_0 = S_0 e^{(r-d)T}$

$F_0 = S_0 e^{(r-r^f)T}$

$F_0 = (S_0 + U) e^{rT}$

$F_0 = S_0 e^{(r+u)T}$

$F_0 = S_0 e^{(r+u-y)T}$

$f = (F - K) e^{-rT}$

$S_0 \geq c \geq \max\left\{S_0 - D - Ke^{-rT}, 0\right\}$

$Ke^{-rT} \geq p \geq \max\left\{D + Ke^{-rT} - S_0, 0\right\}$

$c + D + Ke^{-rT} = p + S_0$

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ROUGH WORKING
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